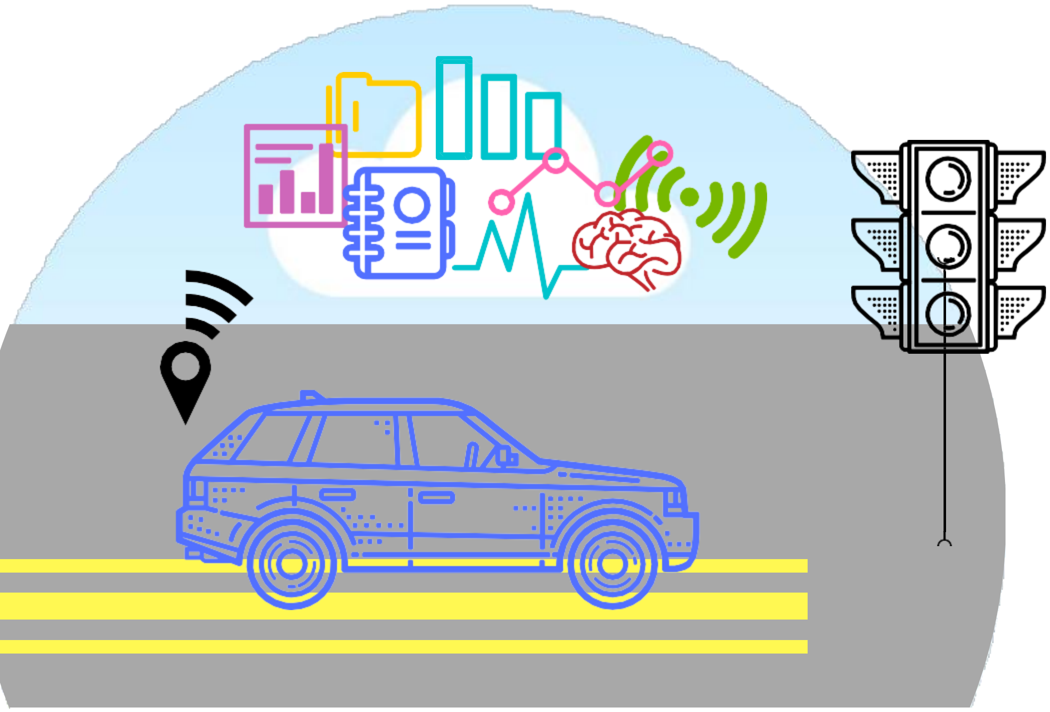


SMART TRAFFIC LIGHT SYSTEM

Tired of the light turning red way too quickly?
Are you stuck waiting on the same intersection for too long?

Nithish Raj(22311A04CE)
Abhinav Kumar(22311A04BU)
Adil Hussain(23315A0448)

60%

of the population are urban residents. More moving vehicles need to be accommodated over a fixed size of transportation infrastructure.

SOURCE: WWW.CNBC.COM

AIM

Control Duration of green light for a specific traffic light at an intersection.

RULE

The traffic signals should not flash the same stretch of green all the time, but should depend on the number of cars present.

BENIFIT 1

ACCIDENTS

Minimize number of accidents (fewer fatalities)
Reduce number of patients in hospitals
Reduce demand on emergency response teams
Release funds of medical supplies

BENIFIT 2

POLLUTION

Minimize carbon footprint
Increase fuel efficiency
Reduce excessive breaking (less wear & tear)

BENIFIT 3

COMMUTE HASTLE AND COST

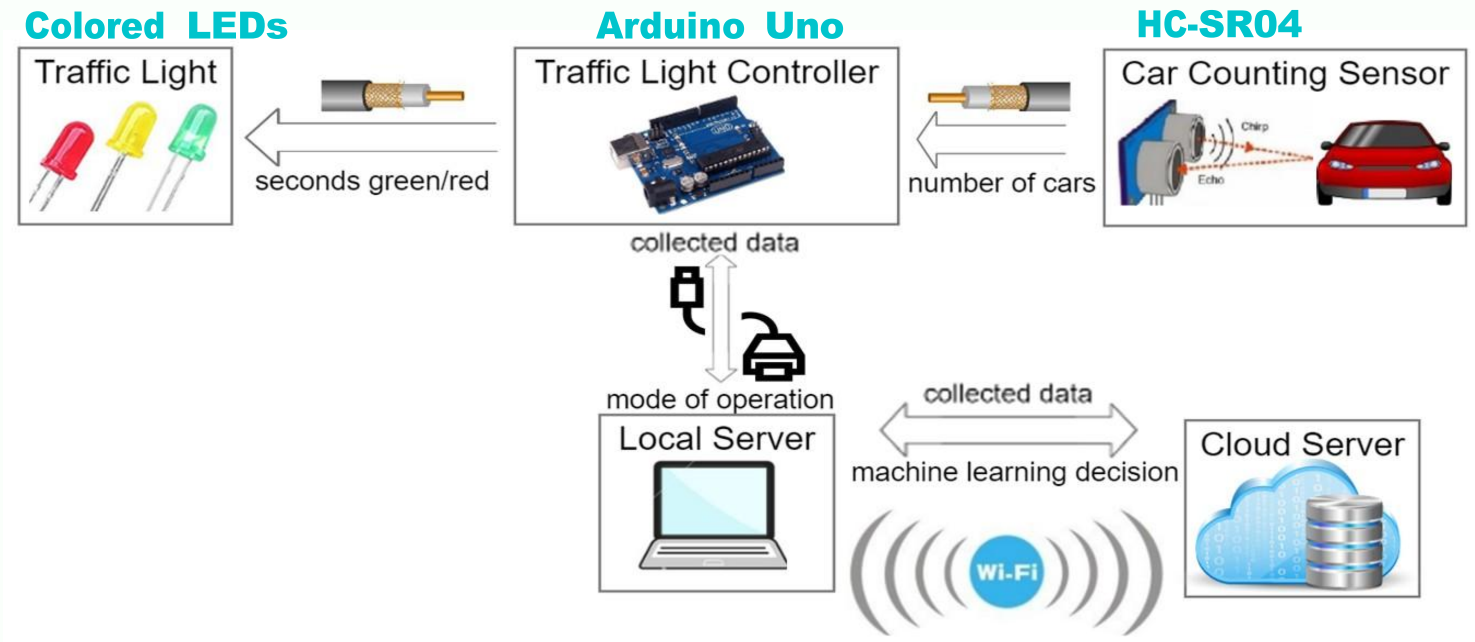
Minimize hassle and cost of commuting
Lower insurance rates (less accidents)
Reduce fuel cost and amount of lost time

USER SCENARIO

Sensor detects increasing flow of traffic

Intelligent System increases number of seconds green light is on

TECHNOLOGY



Colored LEDs
Arduino Uno
HC-SR04
Traffic Light
Traffic Light Controller
Car Counting Sensor
seconds green/red
number of cars
Chirp
Echo
collected data
mode of operation
Local Server
collected data
machine learning decision
Cloud Server
Wi-Fi

WHY TO USE THIS SYSTEM?

LOW BUDGET

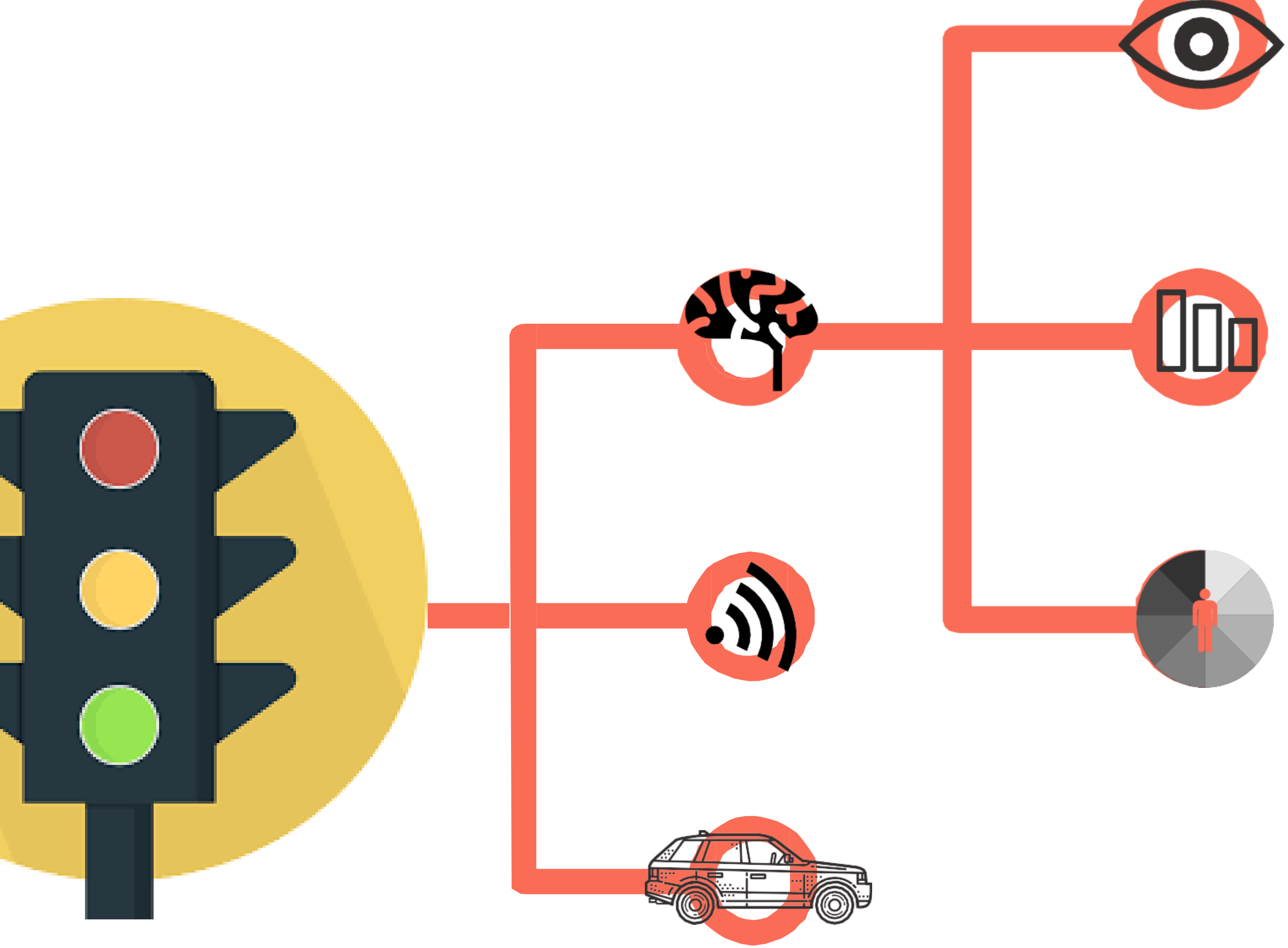
The technology used has a very low price. the system's total price is estimated to be $25

SIMPLE IMPLEMENTATION

The car detection sensor used is non-invasive. No road work is needed to mount the system

EASY MAINTENANCE

Any problem encountered can be solved remotely because the system is connected to the internet



THANK YOU!